# Standard Hourly Forecast
## API User Document
### Version 1.0



## Table of Contents

- Audience
- Geography
- Background Technology
- Response Format
- Icon Codes, Weather Phrases and Images
- Translations
  - Fields Translated
- Expiration
- URL Construction
- Overview
- Unit of Measure Requirement
- URL Format
- Data Elements & Rule Definitions
- Response Field Maintenance
- Formatted Response Sample
  - XML Example
  - JSON Example
- Display Examples

## Audience
This API is intended for web and mobile platforms.

## Geography
Worldwide.

## Background Technology
This API is a REST-based web service.

## Response Format
This TWC API can return either JSON or XML formatted responses.

## Icon Codes, Weather Phrases and Images
For the mapping of icon codes, weather phrases and images please refer to the Icon Code, Weather Phrases and Images document.

## Translations
This TWC API handles the translation of phrases. However, when formatting a request URL a valid language must be passed along (see the language code table for the supported codes).

### Fields Translated

- golf_category
- wdir_cardinal
- phrase_32char
- dow
- uv_desc

## Data Lifetime - Caching & Expiration
Standard HTTP Cache-Control headers are used to define caching length. The TTL value is provided in the HTTP Header as an absolute time value using the “Expires” parameter, for example: “Expires: Fri, 12 Jul 2013 12:00:00 GMT” The response provides a data element expire_time_gmt. The value in this data element should be used to expire and remove a record from your system.

## URL Construction
Please refer to the TWC API Common Usage document for a tutorial on URL construction and URL references.

## Overview

The  Standard Hourly Forecast  API is sourced from the TWC Forecast system.  This TWC API returns weather forecasts for the current day up to 2 days out. Your content licensing agreement with TWC determines the number of days returned in the API response and is constrained by the API Key that is provided to your company. Please refer to the Data Elements section later in this document for more details.

> Your content licensing agreement with TWC determines the number of days returned in the API response and is constrained by the API Key entitlements that is provided to your company. Each days or hours duration (6,12,24,48 hours) is an atomic API endpoint. Your API key must be authorized for each atomic API endpoint to successfully request a given API endpoint. If your API key is authorized for only the 48 hour endpoint, and you attempt to request a 24 hours duration, you will get an error stating that "Api not allowed for this api key."

## Unit of Measure Requirement

The unit of measure for the response. The following values are supported:

- e = English units
- m = Metric units
- h = Hybrid units (UK)
- s  = Metric SI units (not available for all APIs)

## URL Format

| Atomic API URL Examples:<br>Your content licensing agreement with TWC determines the number of days returned in the API response and is constrained by the API Key that is provided to your company. | |
|---|---|
| **Request by Geocode (Latitude & Longitude):**<br>https://api.weather.com/v1/**geocode/34.063/-84.217**/forecast/hourly/24hour.**json?language=en-US**&**units=e**&**apiKey=yourApiKey** | **Required Parameters:**<br>geocode, language, format |
| https://api.weather.com/v1/geocode/34.063/-84.217/**forecast/hourly/6hour**.json?language=en-US&units=e&apiKey=yourApiKey<br>https://api.weather.com/v1/geocode/34.063/-84.217/**forecast/hourly/12hour**.json?language=en-US&units=e&apiKey=yourApiKey<br>https://api.weather.com/v1/geocode/34.063/-84.217/**forecast/hourly/24hour**.json?language=en-US&units=e&apiKey=yourApiKey<br>https://api.weather.com/v1/geocode/34.063/-84.217/**forecast/hourly/48hour**.json?language=en-US&units=e&apiKey=yourApiKey | |
| **Request by Postal Code:**<br>The Postal Code has a TWC proprietary location type (4) with the following format: location/<postal code>:<location type>:<country code><br>https://api.weather.com/v1/**location/30075:4:US**/forecast/hourly/24hour.**json?language=en-US**&**units=e**&**apiKey=yourApiKey** | **Required Parameters:**<br>postal code:4:country code, language, format |
| https://api.weather.com/v1/location/30075:4:US/**forecast/hourly/6hour**.json?language=en-US&units=e&apiKey=yourApiKey<br>https://api.weather.com/v1/location/30075:4:US/**forecast/hourly/12hour**.json?language=en-US&units=e&apiKey=yourApiKey<br>https://api.weather.com/v1/location/30075:4:US/**forecast/hourly/24hour**.json?language=en-US&units=e&apiKey=yourApiKey<br>https://api.weather.com/v1/location/30075:4:US/**forecast/hourly/48hour**.json?language=en-US&units=e&apiKey=yourApiKey | |

## Data Elements & Rule Definitions

Each data element has three rules associated with it as defined below.

| This Rule ... | ... does this ... | ... answers this ... |
|---|---|---|
| **Usage Rule** | Determines whether a data element is required or optional.<br>If it is optional, determines whether or not you can substitute it with a different data element. | Must I use this data element or can I replace it with a different one? |
| **Processing Rule** | Defines how to process a data element so the results are correct. | If I use this data element, how do I process it? |
| **Display Rule** | Defines the proper display format for a data element. | How do I display this data element? |

## Data Element Descriptions

| Outbound JSON/XML | Description | Type | Length | Range | Null | Sample | Usage | Processing | Display |
|---|---|---|---|---|---|---|---|---|---|
| **Metadata** | | | | | | | | | |
| Metadata fields including echo parameters defined in API Common Usage & Style Guide | | | | | | | | | |
| **Hourly Forecast** | **This section will repeat 24 times** | | | | | | | | |
| class | Data identifier | string | | | N | fod_long_range_hourly | required | none | do not display |
| expire_time_gmt | Expiration time in UNIX seconds | epoch | 11 | | N | 1373914800 | required | none | do not display |
| fcst_valid | Time forecast is valid in UNIX seconds | epoch | 10 | | N | 1369306800 | required | none | do not display |
| fcst_valid_local | Time forecast is valid in local apparent time. | ISO | | | N | 2013-08-06T07:00:00-0400 | required | none | do not display |
| num | This data field is the sequential number that identifies each of the forecasted days in the API. They start on day 1, which is the forecast for the current day. Then the forecast for tomorrow uses number 2, then number 3 for the day after tomorrow, and so forth. | Integer | 2 | 1 - 15 | N | 1 | optional | none | display as provided |
| day_ind | This data field indicates whether it is daytime or nighttime based on the Local Apparent Time of the location. | string | 1 | D = Day,<br>N = Night,<br>X = missing (for extreme northern and southern hemisphere | N | D | optional | none | do not display |
| temp | The temperature of the air, measured by a thermometer 1.5 meters (4.5 feet) above the ground that is shaded from the other elements. You will receive this data field in Fahrenheit degrees or Celsius degrees. | integer | 4 | -140 to 140 (F) | N | 68 | required | none | Display as provided in degrees Fahrenheit or degrees Celsius based on the Unit of Measure in the API request. Always display the unit of temperature (°F or °C) with the value. |
| dewpt | The temperature which air must be cooled at constant pressure to reach saturation. The Dew Point is also an | integer | 3 | -80 to 100 (°F) or -62 to 37 (°C) | N | 63 | optional | none | Display as provided in degrees Fahrenheit or degrees Celsius based on the Unit of Measure in the API request. Always |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | indirect measure of the humidity of the air. The Dew Point will never exceed the Temperature. When the Dew Point and Temperature are equal, clouds or fog will typically form. The closer the values of Temperature and Dew Point, the higher the relative humidity. | | | | | | | | display the unit of temperature (°F or °C) with the value. |
| hi | Hourly maximum heat index.<br>An apparent temperature. It represents what the air temperature "feels like" on exposed human skin due to the combined effect of warm temperatures and high humidity.<br>When the temperature is ***70°F or higher***, the Feels Like value represents the computed **Heat Index**.<br>For temperatures ***between 40°F and 70°F***, the Feels Like value and Temperature are the same, regardless of wind speed and humidity, so use the Temperature value. | integer | 4 | | Y | 84 | optional | Display Heat Index only when the Heat Index value in your data feed is more than 21°C or 70°F. | Use either Celsius degrees **or** Fahrenheit degrees **or** both. Always display the unit of temperature (°F or °C) with the value. |
| wc | Hourly minimum wind chill.<br>An apparent temperature. It represents what the air temperature "feels like" on exposed human skin due to the combined effect of the cold temperatures and wind speed.<br>When the temperature is ***61°F or lower*** the Feels Like value represents the computed **Wind Chill** so display the Wind Chill value.<br>For temperatures ***between 61°F and 75°F***, the Feels Like value and Temperature are the same, regardless of wind speed and humidity, so display the Temperature value. | integer | 4 | | N | 68 | optional | Display Wind Chill only when the Wind Chill value in your data feed is less than 5°C or 40°F. | Use either Celsius degrees **or** Fahrenheit degrees **or** both. Always display the unit of temperature (°F or °C) with the value. |
| feels_like | Hourly feels like temperature.<br>An apparent temperature. It represents what the air temperature "feels like" on exposed human skin due to the combined effect of the wind chill or heat index. | integer | 4 | | N | 84 | optional | none | When the temperature is ***40°F or lower*** the Feels Like value represents the computed **Wind Chill** so display the Wind Chill value.<br>When the temperature is ***70°F or higher***, the Feels Like value represents the computed **Heat Index** so display the Heat Index value. |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | For temperatures ***between 40°F and 70°F***, the Feels Like value and Temperature are the same, regardless of wind speed and humidity, so display the Temperature value.<br>Always display the unit of temperature (°F or °C) with the value. |
| icon_extd | Code representing explicit full set sensible weather. Please refer to the [Forecast Icon Code, Weather Phrases and Images]() document. | integer | 4 | | N | 5500 | required | none | do not display |
| wxman | Code combining Hourly sensible weather and temperature conditions | string | 6 | | N | wx4400 | | | |
| icon_code | This number is the key to the weather icon lookup. The data field shows the icon number that is matched to represent the observed weather conditions. Please refer to the [Forecast Icon Code, Weather Phrases and Images]() document. | integer | 2 | | N | 26 | required | none | do not display |
| dow | Day of week | string | 10 | | N | Thursday | required | You must display this field in your application. According to the space limits of your application to show text, use the name of the week in its abbreviated form.<br>**Examples**:<br>Monday MON Mon. M<br>Tuesday TUE Tue. Tu<br>Wednesday WED Wed. W<br>Thursday THU Thur. Th<br>Friday FRI Fri. F<br>Saturday SAT Sat. Sa<br>Sunday SUN Sun. Su | display as processed by your system |
| phrase_12char | Hourly sensible weather phrase | string | 12 | | N | Cloudy | required | none | display as provided |
| phrase_22char | Hourly sensible weather phrase | string | 22 | | N | Cloudy | required | none | display as provided |
| phrase_32char | Hourly sensible weather phrase | string | 32 | | N | Fog Late | required | none | display as provided |
| subphrase_pt1 | Part 1 of 3-part hourly sensible weather phrase | string | | | N | Cloudy | optional | none | The three parts are to be displayed one after another in numerical order.<br>display as provided |
| subphrase_pt2 | Part 2 of 3-part hourly sensible weather phrase | string | 9 | | N | Late | optional | none | The three parts are to be displayed one after another in numerical order.<br>display as provided |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| subphrase_pt3 | Part 3 of 3-part hourly sensible weather phrase | string | 9 | | N | Thunder | optional | none | The three parts are to be displayed one after another in numerical order.<br>display as provided |
| pop | Hourly maximum probability of precipitation | integer | 3 | 0 to 100 | N | 20 | required | none | Display the percent **%** sign after the value |
| precip_type | The short text describing the expected type accumulation associated with the Probability of Precipitation (POP) display for the hour. | string | 6 | rain,snow, precip | N | rain | required | none | display as provided |
| qpf | The forecasted measurable precipitation (liquid or liquid equivalent) during the hour. | decimal | 5,2 | | N | 0.06 | optional | none | Display as provided with the correct unit of measure (inches or millimeters). |
| snow_qpf | The forecasted hourly snow accumulation during the hour. | decimal | 5,1 | | N | 0 | optional | none | Display as provided with the correct unit of measure (inches or centimeters). |
| rh | The relative humidity of the air, which is defined as the ratio of the amount of water vapor in the air to the amount of vapor required to bring the air to saturation at a constant temperature. Relative humidity is always expressed as a percentage. | integer | 3 | 0 to 100 | N | 83 | required | none | You must display the percent sign “%” after the value. |
| wspd | The maximum forecasted hourly wind speed.<br>The wind is treated as a vector; hence, winds must have direction and magnitude (speed). The wind information reported in the hourly current conditions corresponds to a 10-minute average called the sustained wind speed. Sudden or brief variations in the wind speed are known as “wind gusts” and are reported in a separate data field.<br>Wind directions are always expressed as "from whence the wind blows" meaning that a North wind blows from North to South. If you face North in a North wind the wind is at your face. Face southward and the North wind is at your back. | integer | 3 | | N | 5 | required | none | Display the Wind Speed with its Wind Direction. Use the value as it appears in the data feed (numeric value) and always display its unit of measure, either the fully spelled version or its abbreviation.<br>**Examples**<br>***Wind:*** *from the Southeast at 8 miles per hour.*<br>***Wind:*** *from the Northwest at 12 kilometers/hour.* |
| wdir | Hourly average wind direction in magnetic notation. | integer | 3 | 0 to 359 | N | 145 | required | none | Display the Wind Speed with its Wind Direction. Use the value as it appears in the data feed (numeric value) and always display its unit of measure, either the fully spelled version or its abbreviation.<br>**Examples**<br>***Wind:*** *from the Southeast at 8* |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | *miles per hour.*<br>***Wind:*** *from the Northwest at 12 kilometers/hour.* |
| wdir_cardinal | Hourly average wind direction in cardinal notation. | string | 4 | N , NNE , NE, ENE, E, ESE, SE, SSE, S, SSW, SW, WSW, W, WNW, NW, NNW, CALM, VAR | N | SE | required | none | Display the Wind Speed with its Wind Direction. Use the value as it appears in the data feed (numeric value) and always display its unit of measure, either the fully spelled version or its abbreviation.<br>**Examples**<br>***Wind:*** *from the Southeast at 8 miles per hour.*<br>***Wind:*** *from the Northwest at 12 kilometers/hour.* |
| gust | The maximum expected wind gust speed. | integer | 3 | | Y | 7 | required | none | It is a required display field if Wind Speed is shown. The speed of the gust can be expressed in miles per hour or kilometers per hour. |
| clds | Hourly average cloud cover expressed as a percentage. | integer | 3 | 0 to 100 | N | 82 | optional | none | You must display the percent sign “%” after the value. |
| vis | Prevailing hourly visibility | decimal | 6,3 | 0 to 999 | N | 5.2 | optional | none | display as provided;<br>For value > 1 = no decimal.<br>For value <1 = 2 decimal places |
| mslp | Hourly mean sea level pressure | decimal | 5,2 | | N | 30.21 | required | none | display as provided |
| uv_index_raw | The non-truncated UV Index which is the intensity of the solar radiation based on a number of factors. | decimal | 4,2 | | N | 2.22 | optional | none | do not display |
| uv_index | Hourly maximum UV index | integer | 3 | | N | 2 | optional | None. | Display as provided. If the data value is greater than or equal to 11, convert the value to "10+" |
| uv_desc | The UV Index Description which complements the UV Index value by providing an associated level of risk of skin damage due to exposure. | string | 20 | -2 is Not Available<br>-1 is No Report<br>0 to 2 is Low<br>3 to 5 is Moderate<br>6 to 7 is High<br>8 to 10 is Very High<br>11 to 16 is Extreme | N | Low | optional | none | display as provided |
| uv_warning | TWC-created UV warning based on UV index of 11 or greater. | integer | 1 | | N | 0 | optional | If the data value is 1, then a UV warning is in | do not display |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | effect. If the data value is 0, then no UV warning is in effect. | |
| golf_index | The Golf Index expresses on a scale of 0 to 10 the weather conditions for playing golf. Not applicable at night. | integer | 2 | | Y | 8 | optional | none | display as provided |
| golf_category | The Golf Index Category expressed as a worded phrase the weather conditions for playing golf. | string | 20 | | Y | Very Good | optional | none | display as provided |
| severity | A number denoting how impactful is the forecasted weather for this hour. Can be used to determine the graphical treatment of the weather display such as using red font on weather.com. | integer | 1 | 0 = no threat<br>6 = dangerous / life threatening | N | 2 | optional | none | do not display. for TWC internal use only. |

## Response Field Maintenance

TWC strives to minimize the impact of changes in our weather content to your applications. TWC will not remove, rename or change the data type (int, string) of any data fields in the API response. However, TWC may add new data fields without notice.

**Note:** Outbound File Format: If data is null, then the data element tag will be displayed with the value “null” If the data value is an empty string, the element tag will return the tag and the value will have no value displayed (XML) or display double quotes with no data (JSON).

## Formatted Response Sample

**XML Example**


```xml
<document xmlns="">
   <metadata>
      <language>en-US</language>
      <transaction_id>1427747110973:-323244176</transaction_id>
      <version>1</version>
      <latitude>34.06</latitude>
      <longitude>-84.21</longitude>
      <units>m</units>
      <expire_time_gmt>1427747595</expire_time_gmt>
   </metadata>
   <forecasts>
      <forecast>
         <class>fod_short_range_hourly</class>
         <expire_time_gmt>1427747595</expire_time_gmt>
         <fcst_valid>1427749200</fcst_valid>
         <fcst_valid_local>2015-03-30T17:00:00-0400</fcst_valid_local>
         <num>1</num>
```


**JSON Example**

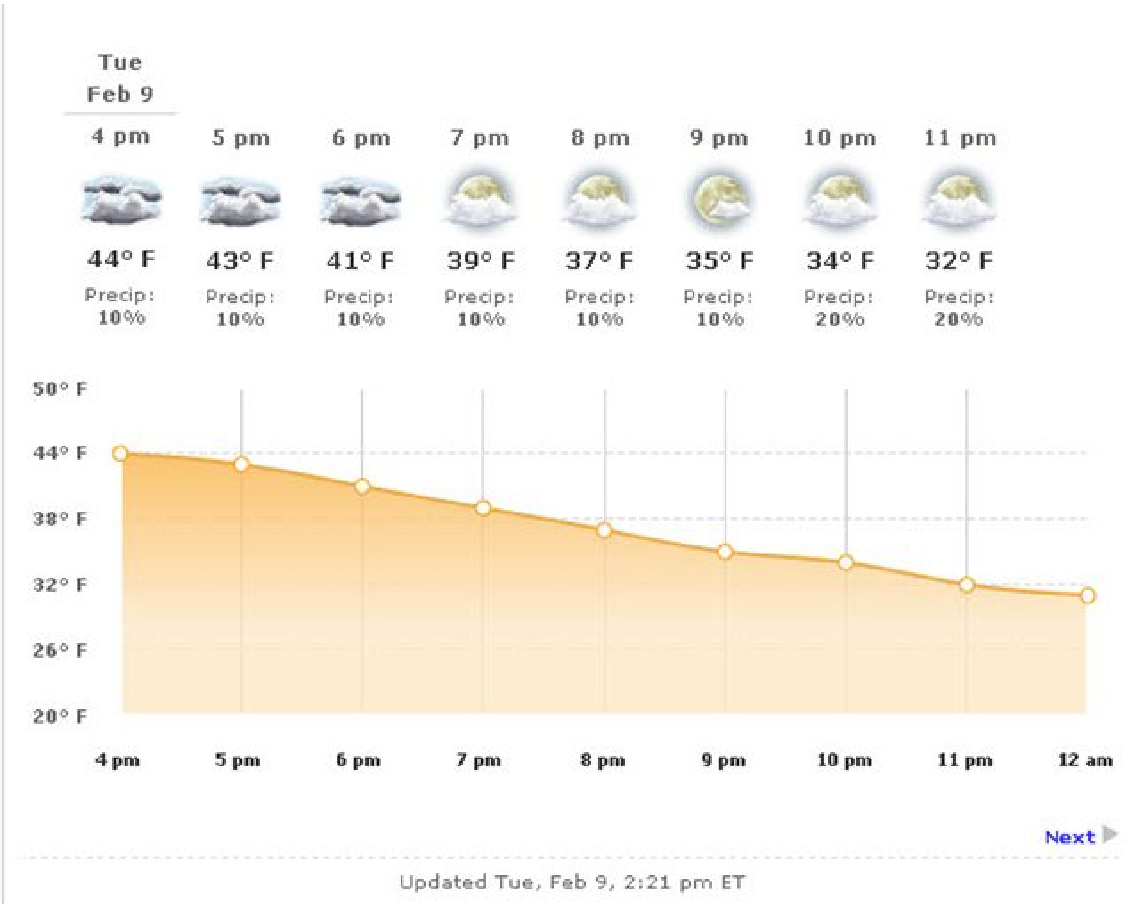

```json
{
"metadata": {
"language": "en-US",
"transaction_id": "1427746995439:365513300",
"version": "1",
"latitude": 34.06,
"longitude": -84.21,
"units": "m",
"expire_time_gmt": 1427747235
},
"forecasts": [
{
"class": "fod_long_range_hourly",
"expire_time_gmt": 1427747048,
"fcst_valid": 1427745600,
"fcst_valid_local": "2015-03-30T16:00:00-0400",
"num": 1,
```

```
        <day_ind>D</day_ind>
        <temp>20</temp>
        <dewpt>5</dewpt>
        <hi>20</hi>
        <wc>20</wc>
        <feels_like>20</feels_like>
        <icon_extd>3400</icon_extd>
        <wxman>wx1000</wxman>
        <icon_code>34</icon_code>
        <dow>Monday</dow>
        <phrase_12char>M Sunny</phrase_12char>
        <phrase_22char>Mostly Sunny</phrase_22char>
        <phrase_32char>Mostly Sunny</phrase_32char>
        <subphrase_pt1>Mostly</subphrase_pt1>
        <subphrase_pt2>Sunny</subphrase_pt2>
        <subphrase_pt3/>
        <pop>0</pop>
        <precip_type>rain</precip_type>
        <qpf>0.0</qpf>
        <snow_qpf>0.0</snow_qpf>
        <rh>39</rh>
        <wspd>20</wspd>
        <wdir>300</wdir>
        <wdir_cardinal>WNW</wdir_cardinal>
        <gust>28</gust>
        <clds>25</clds>
        <vis>16.1</vis>
        <mslp>1018.68</mslp>
        <uv_index_raw>2.76</uv_index_raw>
        <uv_index>3</uv_index>
        <uv_warning>0</uv_warning>
        <uv_desc>Moderate</uv_desc>
        <golf_index>10</golf_index>
        <golf_category>Excellent</golf_category>
        <severity>1</severity>
      </forecast>
      //This API will repeat up to an additional 24 times per response
      {}, // - Response Repeats for Day 2
      {}, // - Response Repeats for Day 3...
      {} // - Response Repeats for Day 24
    </forecasts>
</document>
```

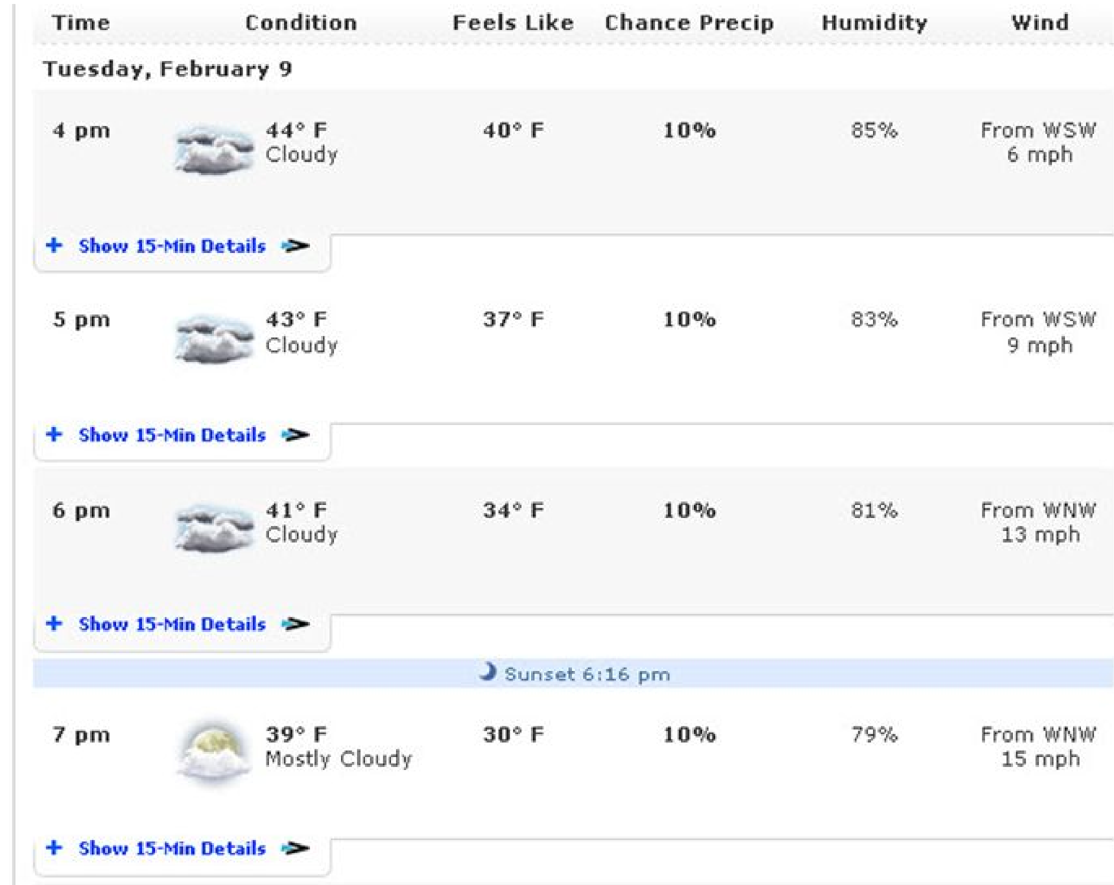
```
"day_ind": "D",
"temp": 20,
"dewpt": 8,
"hi": 20,
"wc": 20,
"feels_like": 20,
"icon_extd": 3400,
"wxman": "wx1000",
"icon_code": 34,
"dow": "Monday",
"phrase_12char": "M Sunny",
"phrase_22char": "Mostly Sunny",
"phrase_32char": "Mostly Sunny",
"subphrase_pt1": "Mostly",
"subphrase_pt2": "Sunny",
"subphrase_pt3": "",
"pop": 0,
"precip_type": "rain",
"qpf": 0,
"snow_qpf": 0,
"rh": 46,
"wspd": 19,
"wdir": 302,
"wdir_cardinal": "WNW",
"gust": 27,
"clds": 23,
"vis": 16.1,
"mslp": 1019.2,
"uv_index_raw": 4.8,
"uv_index": 5,
"uv_warning": 0,
"uv_desc": "Moderate",
"golf_index": 9,
"golf_category": "Very Good",
"severity": 1
},
//This API will repeat up to an additional 24 times per response
{}, // - Response Repeats for Day 2
{}, // - Response Repeats for Day 3...
{} // - Response Repeats for Day 24
]
}
```

## Display Examples

**Example 1**




**Example 2**



| Time | Condition | Feels Like | Chance Precip | Humidity | Wind |
|---|---|---|---|---|---|
| Tuesday, February 9 | | | | | |
| 4 pm | 44° F Cloudy | 40° F | 10% | 85% | From WSW 6 mph |
| + Show 15-Min Details | | | | | |
| 5 pm | 43° F Cloudy | 37° F | 10% | 83% | From WSW 9 mph |
| + Show 15-Min Details | | | | | |
| 6 pm | 41° F Cloudy | 34° F | 10% | 81% | From WNW 13 mph |
| + Show 15-Min Details | | | | | |
| Sunset 6:16 pm | | | | | |
| 7 pm | 39° F Mostly Cloudy | 30° F | 10% | 79% | From WNW 15 mph |
| + Show 15-Min Details | | | | | |